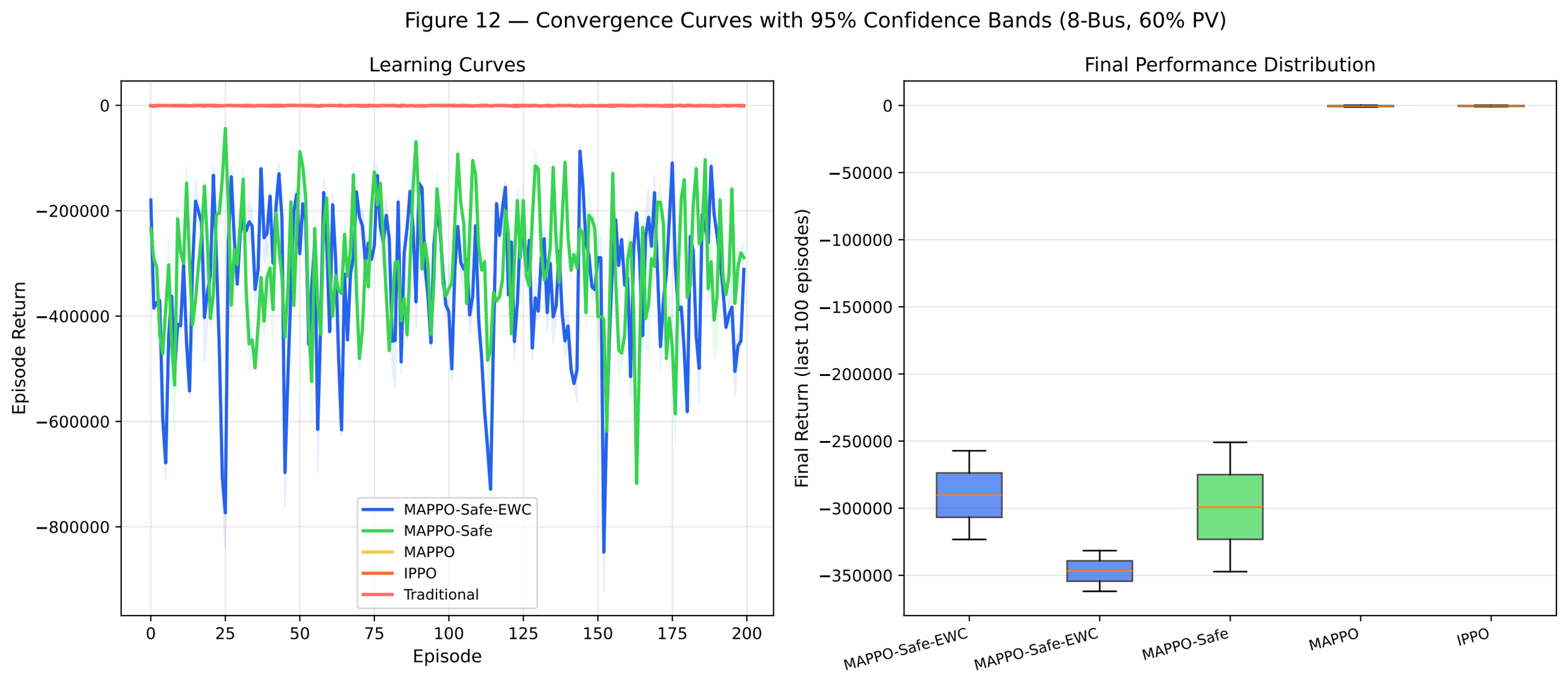
Figure 12 — Convergence Curves with 95% Confidence Bands (8-Bus, 60% PV)
Learning Curves
Episode Return
Episode
0
−200000
−400000
−600000
−800000
0
25
50
75
100
125
150
175
200
MAPPO-Safe-EWC
MAPPO-Safe
MAPPO
IPPO
Traditional
Final Performance Distribution
Final Return (last 100 episodes)
0
−50000
−100000
−150000
−200000
−250000
−300000
−350000
MAPPO-Safe-EWC
MAPPO-Safe-EWC
MAPPO-Safe
MAPPO
IPPO